TED(21)-5303C **REVISION 2021**

MODEL QUESTION PAPER

EMBEDDED AND IOT SYSTEM DESIGN-SET-I

(Maximum Marks : 75) **TIME : 3HRS**

PART -A

Answer all questions in one word or one sentence

I.

1. Define an Embedded system?
2. List any two application of Embedded systems?
3. Explain any two features of 8051Microcontroller?
4. What is the function of Program Counter in 8051?
5. What you meant by IoT?
6. List any two requirements of IoT?
7. What you meant by M2M?
8. Define the use of Webserver in IoT?
9. What is the function of REST in IoT?

(9x1=9)

PART B

Answer any eight questions from the following. Each question carries 3 marks.

II.

1. What are the characteristics of an embedded system?
2. Enumerate essential functional blocks of an embedded system?
3. Describe any one serial communication standards used in embedded systems?
4. Explain the various addressing modes of 8051 with proper examples?
5. Discuss the bit pattern for 8051 flag register (PSW) ?
6. Describe IoT enabling technologies?
7. Explain wireless sensor network?
8. Discuss any two IoT communication APIs?
9. Differentiate IoT and M2M?
10. Discuss about the need for IP optimization in IoTs?

(8x3=24)

PART C

Answer all questions from the following. Each question carries 7 marks.

III

1. Explain in detail the different steps involved in the embedded system design process with an example and neat diagram?

OR

2. Compare serial communication with parallel communication. Draw the diagram of I2C frame format. Explain each field?

3. Draw the architecture and also explain the functions of each block 8051 microcontroller in detail?

OR

4. Explain the timers of 8051 Microcontroller and their modes of operation in 8051?

5. With a neat schematic explain the interfacing of stepper motor with 8051 Microcontroller?

OR

6. Explain the interfacing of seven segment display to 8051 Microcontroller?

7. With the help of a diagram explain the basic building blocks of an IoT architecture?

OR

8. Discuss about Zigbee protocol architecture with a neat figure?

9. Describe on physical design of IoT?

OR

10. Discuss about any three types of WSN topologies used in IoT ?

11. Discuss about the concept of M2M in detail with a neat diagram?

OR

12. Explain the different types of IoT protocols used in different layers?

(6x7=42)

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MODEL QUESTION PAPER

EMBEDDED AND IOT SYSTEM DESIGN-SET-2

(Maximum Marks : 75) **TIME : 3HRS**

PART -A

Answer all questions in one word or one sentence

I.

1. List any two challenges of Embedded systems?
2. What are the major components of Embedded systems.
3. Define Data Pointer in 8051?
4. What are the interrupts in 8051?
5. What you meant by RFID?
6. Define WSN topology?
7. What is the main role of M2M in IoT?
8. What you meant by COAP?
9. What is the main use of HTTP in IoT?

(9x1=9)

PART B

Answer any eight questions from the following. Each question carries 3 marks.

II.

1. Differentiate Embedded Systems and general computing systems?
2. Write a short note on characteristics of Embedded system.
3. **What are some applications of an embedded system?**
4. What is an interrupt?
5. List some features of 8051 Microcontroller.
6. Define **Cloud Computing and its characteristics.**
7. **Compare Zigbee with WIFI.**
8. What are different types of sensors in IoT?
9. What is the use of 6LoWPAN in the IOT?
10. What is MQTT?

(8x3=24)

PART C

Answer all questions from the following. Each question carries 7 marks.

III

1. Describe embedded system with its classification.

OR

2. Explain embedded system design with block diagram.

3. Draw internal block schematic of 8051 and explain the architecture.

OR

4. Explain addressing modes of 8051 in detail.

5. Discuss about the memory structure of 8051with neat figure.

OR

6. Write a program to multiply two 8 – bit numbers using 8051.

7. Explain the architecture of IoT in detail

OR

8. What are the types of RFID systems? Explain in detail.

9. Describe on GSM channel types.

OR

10. Explain IoT platform and what are the types of IoT platforms?

11. Discuss about four types of IoT networks in detail.

OR

12. With the help of a block diagram explainMQTT architecture.

(6x7=42)